

Project Report: Themis's Domain

Advanced Crime Record & Management System (CRMS)

Bengaluru Police Department Command & Analytics Platform

*[!NOTE] This document serves as a comprehensive, production-grade technical report outlining the architecture, system design, feature set, security controls, database schemes, and data flows of **Themis's Domain** (formerly Themis Nomos / HeraRecord). This report is structured for academic project submissions and production engineering reviews.*

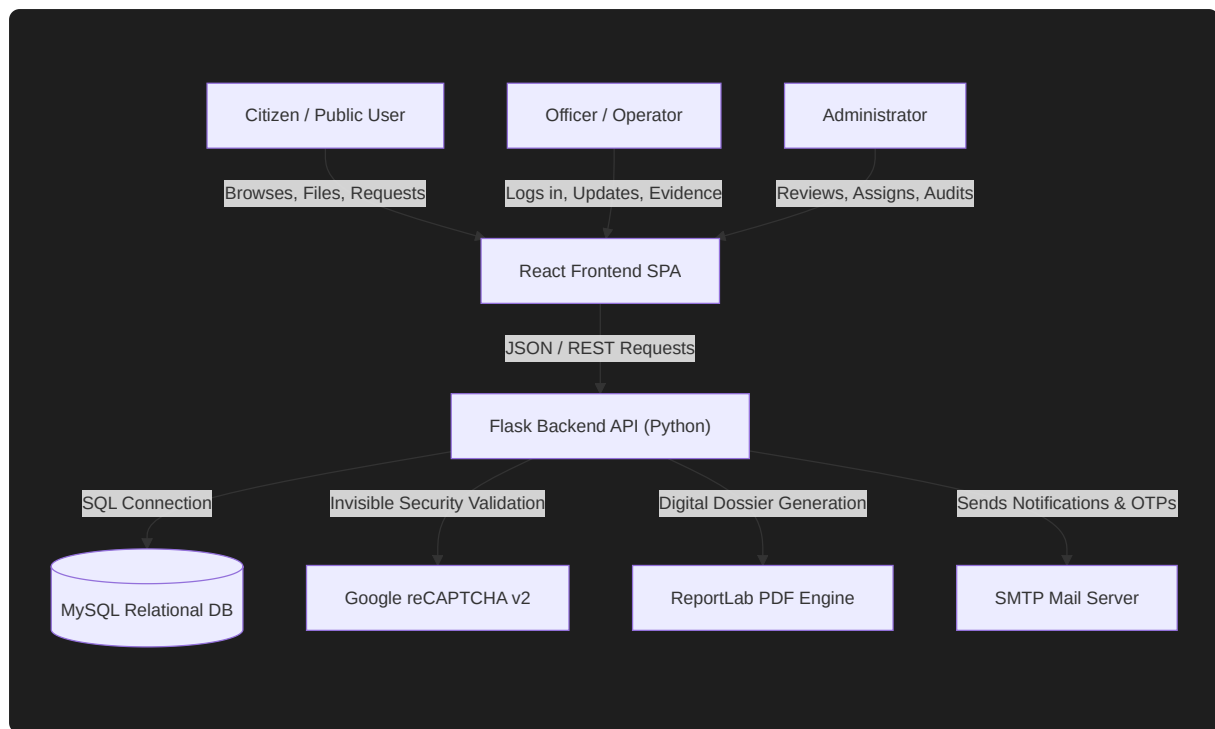
1. Executive Summary

Themis's Domain is a highly polished, responsive, and secure digital command platform designed for the **Bengaluru Police Department**. The system replaces manual, error-prone crime tracking operations with an automated, role-based case management and citizen reporting platform.

It provides an end-to-end framework starting from public complaint intake, transitioning into automated workload-balanced officer case assignments, proceeding to evidentiary record gathering, and concluding with legally compliant case dossier generation and digital signature verification.

2. System Architecture

The application is built using a modern **decoupled client-server architecture**:



Technology Stack

- **Frontend Layer:** Single Page Application (SPA) utilizing React (embedded dynamically inside standard HTML5), styled using custom Vanilla CSS alongside a glassmorphism utility design system, and powered by Framer Motion for premium micro-animations.
- **Backend API Layer:** Python-based Flask server executing a RESTful API pattern, incorporating thread-safe local stores, secure file system ingestion, and connection pooling.
- **Database Layer:** MySQL database utilizing relational indexing, cascading foreign keys, transaction constraints, and multi-value unique constraints.
- **Security Integrations:** Invisible Google reCAPTCHA v2, SHA-256 validation, salt-hashed bcrypt password security, and multi-layer cryptographic validation tokens.
- **Utility Engines:** ReportLab PDF Generator (for dossier document publishing) and secure SMTP mailers (for citizen OTPs, administrative alerts, and officer assignment dispatches).

3. Database Schema Design

The database structure is designed to enforce strict referential integrity. It is structured into **8 core tables**:



Table Definitions

1. officers

Stores police department personnel, access credentials, and administrative roles.

- officer_id** (INT, Primary Key, Auto Increment)
- name** (VARCHAR, Not Null)
- rank** (VARCHAR, Not Null) - e.g., Inspector, Sub-Inspector, Head Constable.
- badge** (VARCHAR, Unique) - e.g., BPD-7821.
- station** (VARCHAR)
- phone** (VARCHAR)
- email** (VARCHAR, Unique)
- join_date** (DATE)
- password_hash** (VARCHAR) - Bcrypt secure hash.

- `role` (ENUM: 'admin', 'inspector', 'viewer')

2. `cases`

The main table storing active and historical crime investigations.

- `case_id` (INT, Primary Key, Auto Increment)
- `title` (VARCHAR, Not Null)
- `description` (TEXT)
- `crime_type` (VARCHAR) - e.g., Cyber Fraud, Theft, Assault, Ponzi Scheme.
- `status` (ENUM: 'Pending Review', 'Recommended', 'Assigned', 'Active', 'Solved', 'Closed', 'Rejected')
- `date_reported` (DATETIME, Default Current Timestamp)
- `location` (VARCHAR)
- `complaint_mode` (ENUM: 'Online', 'Offline')
- `source` (ENUM: 'public', 'officer')
- `complainant_name` (VARCHAR)
- `complainant_contact` (VARCHAR)
- `complainant_aadhaar` (CHAR(12))

3. `case_officer`

A junction table enabling a many-to-many relationship mapping officers to their assigned cases.

- `case_id` (INT, Foreign Key referencing `cases.case_id` ON DELETE CASCADE)
- `officer_id` (INT, Foreign Key referencing `officers.officer_id` ON DELETE CASCADE)
- Primary Key is composite: `(case_id, officer_id)`

4. `public_complaints`

Staging area for citizen-submitted complaints pending triage.

- `complaint_id` (INT, Primary Key, Auto Increment)
- `complainant_name` (VARCHAR, Not Null)
- `contact` (VARCHAR, Not Null)
- `email` (VARCHAR)

- `aadhaar` (CHAR(12), Not Null)
- `crime_type` (VARCHAR, Not Null)
- `location` (VARCHAR, Not Null)
- `incident_desc` (TEXT, Not Null)
- `status` (ENUM: 'Pending', 'Reviewed', 'Promoted', 'Rejected')
- `promoted_case_id` (INT, Nullable, Foreign Key referencing `cases.case_id`)
- `reviewed_by` (INT, Nullable, Foreign Key referencing `officers.officer_id`)
- `reviewed_at` (DATETIME)

5. `case_access_requests`

Manages citizen applications to read secure case dossiers.

- `request_id` (INT, Primary Key, Auto Increment)
- `case_id` (INT, Foreign Key referencing `cases.case_id`)
- `requester_name` (VARCHAR)
- `requester_email` (VARCHAR)
- `requester_number` (VARCHAR)
- `reason` (TEXT)
- `status` (ENUM: 'Pending', 'Accepted', 'Rejected')
- `decided_by` (INT, Foreign Key referencing `officers.officer_id`)
- `decided_at` (DATETIME)

6. `assignment_recommendations`

Stores automated recommendation engine records generated by the workload allocation algorithm.

- `recommendation_id` (INT, Primary Key, Auto Increment)
- `complaint_id` (INT, Foreign Key referencing `public_complaints.complaint_id`)
- `recommended_officer_ids` (JSON) - Array of selected officer IDs.
- `status` (ENUM: 'pending', 'approved', 'rejected')
- `admin_approved_officer_ids` (JSON, Nullable)
- `approved_by` (INT, Foreign Key referencing `officers.officer_id`)
- `approved_at` (DATETIME)

7. `case_updates`

Maintains historical case timeline entries posted by assigned officers.

- `update_id` (INT, Primary Key, Auto Increment)
- `case_id` (INT, Foreign Key referencing `cases.case_id`)
- `officer_id` (INT, Foreign Key referencing `officers.officer_id`)
- `update_text` (TEXT)
- `created_at` (DATETIME)

8. `case_evidence`

Tracks evidentiary files, secure physical paths, and audit details.

- `evidence_id` (INT, Primary Key, Auto Increment)
- `case_id` (INT, Foreign Key referencing `cases.case_id`)
- `officer_id` (INT, Foreign Key referencing `officers.officer_id`)
- `file_name` (VARCHAR) - Generated secure file name on storage.
- `original_name` (VARCHAR) - Sanitized user file name.
- `file_path` (VARCHAR(512))
- `mime_type` (VARCHAR(100))
- `file_size` (INT)
- `description` (VARCHAR)

4. Key Functional Features

Themis's Domain provides customized interfaces across three user classes:

A. Citizen & Public Portal

- **Public Case Browser:** Interactive table to search active or solved public cases in real-time.
- **Secure Complaint Registration:** Digital intake for criminal incidents requiring full validation.
- **Dossier Access Request System:** Enables involved citizens (e.g., victims, defense counsel) to request certified copies of case updates and evidence records.

- **Multiphase Security Triage:** Uses a multi-step check-in (Email Verification → SMS/OTP Simulation → Disclaimer Acceptance) before filing.

B. Staff & Officer Dashboard

- **Case Dossier Manager:** Full access to assigned cases, including historical logs, active team members, and telemetry.
- **Evidence Vault:** Secure upload panel supporting document, image, video, and audio evidence uploads with custom description logs.
- **Timeline Logs:** Allows case investigators to post micro-updates to the case dossier timeline to track active progress.
- **Secure Session Control:** Maintains dynamic routing keeping state across page reloads unless explicitly signing out.

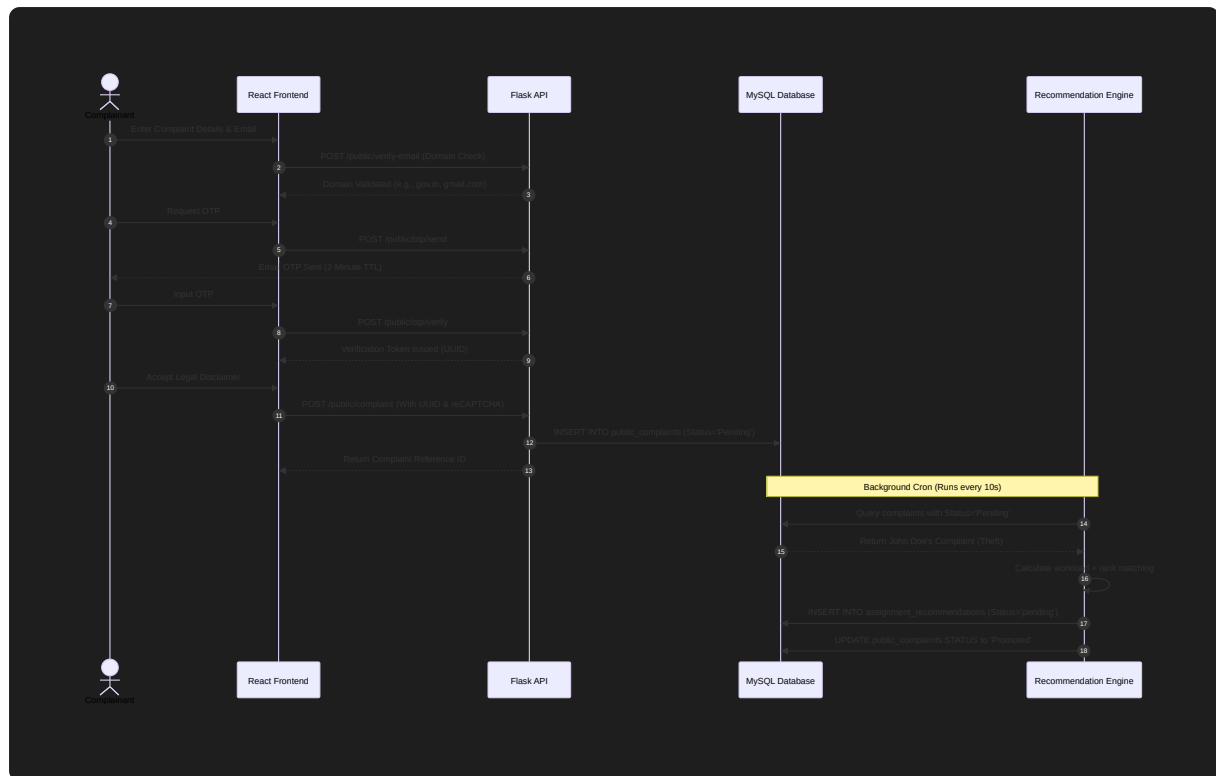
C. Administrator Command Center

- **Triage Queue:** Review citizen-submitted complaints. Direct ability to promote a complaint into an active case, assign officers, or reject false reports.
- **Workload Allocation Console:** Shows active officer workloads, statistics, and assignment maps.
- **Algorithm Review Board:** Interface reviewing system recommendations generated for pending cases, offering the option to approve recommended teams or adjust them manually.

5. Critical Workflows & Data Flows

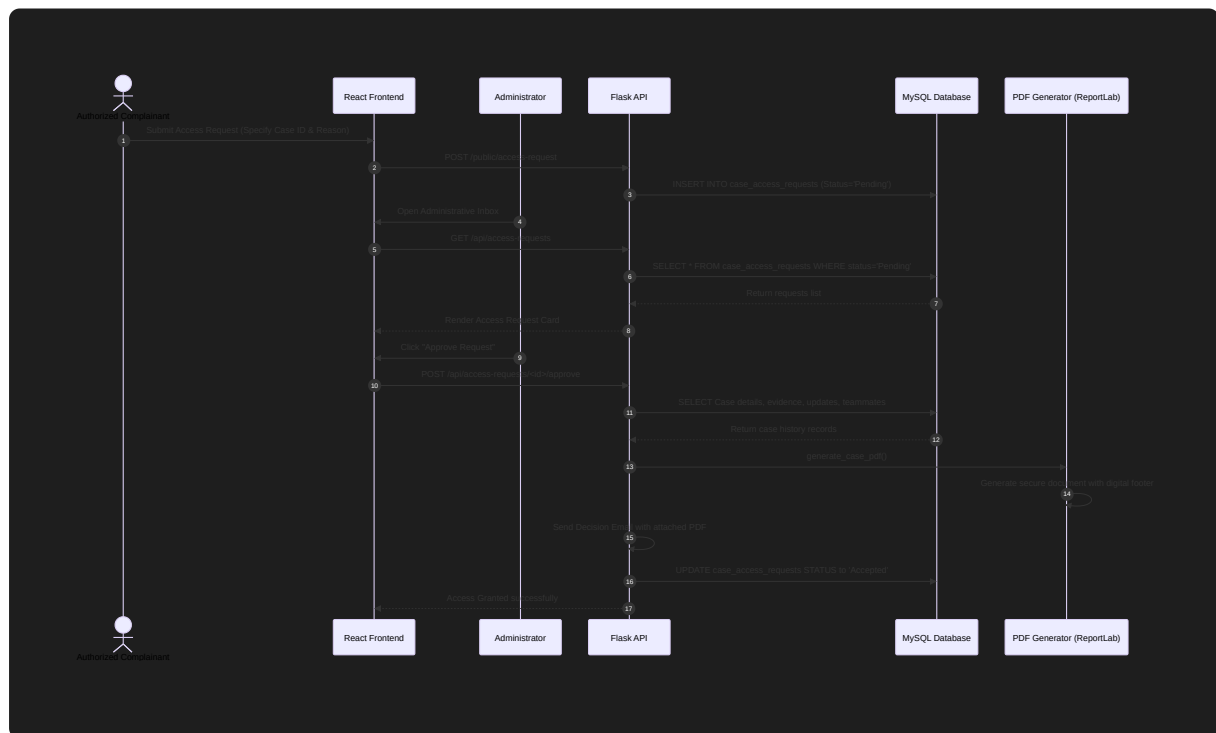
Workflow A: Citizen Complaint Registration & Promotion

This flowchart illustrates the step-by-step process of a citizen filing an online complaint, passing security checks, going through database staging, and getting triaged by the automated assignment engine.



Workflow B: Secure Case Dossier Delivery

This flow outlines how a requester applies for secure document access, which is subsequently reviewed by an administrator and exported as a digitally generated PDF document.



6. Mathematical Model of the Case Assignment Algorithm

The automated allocation engine determines which officers are selected for a case based on **Crime Severity Matching** and a **Minimization Workload Function**.

1. Severity Matching Function

The algorithm maps the reported `crime_type` to a severity level S : $S(\text{crime_type}) \in \{\text{Critical}, \text{High}, \text{Medium}\}$

Based on S , the requirement vector R_S dictates the exact counts of personnel needed per rank:

- Critical: 1 Inspector, 1 Sub-Inspector, 1 Head Constable.
- High: 1 Sub-Inspector, 1 Head Constable.
- Medium: 1 Sub-Inspector, 1 Head Constable.

2. Workload and Seniority Minimization Function

For a required rank r , the algorithm selects available officers by sorting them using a compound lexicographical key. Let the set of officers of rank r be O_r . For each officer $o \in O_r$, we define:

1. **Workload Coefficient (W_o)**: The count of active cases currently assigned to officer o .

$$W_o = \sum_{c \in \text{Cases}} [\text{status}(c) = \text{'Active'} \wedge o \text{ is assigned to } c]$$
2. **Seniority Tiebreaker (J_o)**: The join date of officer o , represented as a UNIX timestamp.

The algorithm sorts O_r in ascending order of the sorting tuple: $\text{SortKey}(o) = \langle W_o, J_o \rangle$

The system assigns the top k officers where k is the requirement count for rank r , effectively minimizing the departmental workload variance: Minimize $\sigma^2(W)$

7. Comprehensive Security Controls

The platform implements multi-layer defense-in-depth:

Security Domain	Implemented Mechanism	Objective / Threat Mitigated
Authentication	Password protection using bcrypt (salt factor = 12).	Prevents credential theft, rainbow table attacks, and unauthorized access.
Access Control	Role-Based Access Control (RBAC) separating roles into <code>admin</code> , <code>inspector</code> , and <code>viewer</code> .	Enforces Least Privilege access to evidence vaults and administrative tools.
Public Security	Invisible Google reCAPTCHA v2 integration on public endpoints.	Prevents denial-of-service (DoS) attacks and bot spamming of complaint channels.
Email Verification	Secure check-in via <code>/public/verify-email</code> filtering out dummy and unverified domains.	Prevents fake email spamming and ensures citizen communication delivery.
OTP Validation	Thread-safe <code>OTPStore</code> with 2-minute expiration and a 3-attempt lock-out .	Prevents brute-forcing of one-time codes and controls request rates.
File System Security	Filename sanitization, size checks, and UUID file storage.	Prevents directory traversal, file overrides, and malicious script execution.
PDF Legality	Cryptographic audit signature stamps printed on generated PDFs.	Ensures document integrity and provides tracking validation for police dossiers.

8. Development Credentials & Testing

For testing, local databases are seeded with standard testing accounts:

- **Administrative Account:** Badge ID: `ADM-0001` · Password: `crms1234`
- **Staff Officer (Inspector):** Badge ID: `BPD-7821` · Password: `crms1234`
- **Staff Officer (Sub-Inspector):** Badge ID: `BPD-6543` · Password: `crms1234`